The UZH protocol: Separating and reducing Gaussian-basis and pseudopotential errors in CP2K

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(Dated: May 29, 2026)

Gaussian-and-plane-wave density-functional theory combines the compactness of atom-centered orbitals with the systematic real-space representation of densities and potentials. Its accuracy, however, is controlled by two coupled approximations: the Gaussian orbital basis and the pseudopotential. We present the UZH protocol, a CP2K-based closed-loop strategy that calibrates molecularly optimized (MOLOPT) Gaussian basis sets on small molecules and validates Goedecker–Teter–Hutter (GTH) separable dual-space pseudopotentials against all-electron full-potential linearized augmented-plane-wave (FP-LAPW) equation-of-state benchmarks using the SIRIUS electronic-structure code. The protocol separates the total error into basis-set and pseudopotential components by comparing CP2K-GTH-UZH, SIRIUS-GTH-UZH, and SIRIUS-FP-LAPW calculations. It extends the Materials Cloud verification strategy to a setting in which the same pseudopotential can be evaluated with a Gaussian basis and with a systematically converged plane-wave representation. This decomposition identifies whether discrepancies originate from the CP2K Gaussian basis, from the pseudopotential itself, or from their combined use. The resulting optimized UZH settings reduce the dominant errors in the original protocol and provide a practical route for improving CP2K calculations with internally generated basis sets and scalar-relativistic GTH pseudopotentials.

I. INTRODUCTION

Density-functional theory is routinely used as a predictive tool in condensed-matter physics, chemistry, and materials science [1, 2]. Reproducibility studies have shown that, once exchange-correlation and physical setup are fixed, residual discrepancies between codes are mostly numerical: basis-set completeness, treatment of the core electrons, integration grids, smearing, and convergence thresholds all matter at the precision level targeted by modern verification workflows [3–6]. A useful protocol therefore has to do more than report agreement with a reference. It should reveal which approximation limits the result and how that approximation can be improved.

This issue is particularly important for CP2K/Quickstep [7, 8], where Kohn-Sham orbitals are represented by atom-centered Gaussian functions and densities are mapped to auxiliary plane-wave grids within the Gaussian-and-plane-wave (GPW) formalism [9]. The approach is efficient for molecules, liquids, interfaces, and extended systems, but its accuracy is governed jointly by the chosen Gaussian basis sets and norm-conserving pseudopotentials. The two are usually chosen together, and their errors are often reported as a single code-to-code deviation. Such a holistic error is useful for users, but it is insufficient for method development because it does not say whether the dominant correction should be a larger or reoptimized basis set, a revised pseudopotential, or both. This is precisely where CP2K offers a distinctive advantage: Quickstep, the SIRIUS interface, and the newly developed ATOM optimization machinery are available within one coherent code ecosystem, allowing basis-set and pseudopotential errors to be analyzed and improved holistically.

Here we formulate and apply what we call the UZH protocol. The protocol combines molecularly optimized (MOLOPT) basis sets [10] with the analytic Goedecker–Teter–Hutter (GTH) family of separable dual-space Gaussian pseudopotentials and its Hartwigsen–Goedecker–Hutter (HGH) relativistic extension [11–13]. Its central idea is to close a loop between molecular calibration and solid-state verification. In the first step, the Gaussian basis is optimized and calibrated against small molecules, following the MOLOPT philosophy of minimizing both energy errors and ill-conditioning of the overlap matrix [10]. In the second step, the resulting CP2K settings are tested in the reproducible unary-crystal workflow of the Automated Interactive Infrastructure and Database for Computational Science (AiiDA) common workflows [5, 6]. In the third step, the total CP2K error is decomposed by using SIRIUS both with the same GTH-UZH pseudopotentials and as a full-potential linearized augmented-plane-wave (FP-LAPW) reference [14–16].

This three-way comparison is the central methodological step. CP2K-GTH-UZH and SIRIUS-GTH-UZH differ in the orbital representation while keeping the same pseudopotential fixed; their difference isolates the Gaussian-basis component. SIRIUS-GTH-UZH and SIRIUS-FP-

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LAPW share the same code infrastructure but differ in the core-electron approximation; their difference isolates the pseudopotential component. The comparison of CP2K-GTH-UZH with SIRIUS-FP-LAPW gives the total practical error of the UZH protocol. Because CP2K also contains the ATOM machinery for basis-set optimization, third-order Douglas–Kroll–Hess (DKH3) atomic calculations [17–19], and GTH pseudopotential fitting, the same framework can diagnose and then improve the associated basis-set and pseudopotential parameter files.

II. THEORY AND PROTOCOL

A. MOLOPT basis sets

MOLOPT basis sets are generally contracted Gaussian basis sets designed for density-functional calculations in gas and condensed phases [10]. For atom A, a contracted basis function in angular-momentum channel l can be written as

$$\chi_{Alm\nu}(\mathbf{r}) = Y_{lm}(\hat{\mathbf{r}}_A) r_A^l \sum_{p=1}^{N_{Al}} c_{Al\nu p} \exp\left(-\alpha_{Alp} r_A^2\right), \quad (1)$$

with $r_A = |\mathbf{r}-\mathbf{R}_A|$ and $\hat{\mathbf{r}}_A = (\mathbf{r}-\mathbf{R}_A)/r_A$, where Y_{lm} is a spherical harmonic, $m = -l, \ldots, l$ is the magnetic quantum number, $p = 1, \ldots, N_{Al}$ enumerates primitive Gaussians in angular-momentum channel l, α_{Alp} are primitive exponents, and ν labels contractions that differ through the coefficients $c_{Al\nu p}$. Valence and polarization flexibility are therefore introduced mainly through the contraction pattern, while the same primitive pool can be kept sufficiently compact for condensed-phase calculations.

The numerical stability of a candidate basis is measured with the molecular overlap matrix

$$S_{\mu\nu}^{(M)} = \langle \chi_\mu | \chi_\nu \rangle_M \quad (2)$$

where the composite indices μ and ν enumerate contracted basis functions of molecule M. For basis level B, the condition number is

$$\kappa_B(M) = \frac{\lambda_{\max}\left[S_B^{(M)}\right]}{\lambda_{\min}\left[S_B^{(M)}\right]}, \quad (3)$$

where $S_B^{(M)}$ is the overlap matrix for basis level B, and $\lambda_{\max}$ and $\lambda_{\min}$ denote its largest and smallest eigenvalues. The MOLOPT optimization target balances accuracy and numerical stability, i.e.

$$\Omega = \sum_{B\in\mathcal{B}} \sum_{M\in\mathcal{T}} \left[\sum_{q\in\mathcal{Q}} w_q \left(\frac{q_{B,M} - q_M^{\mathrm{ref}}}{s_q} \right)^2 + \gamma \ln \kappa_B(M) \right], \quad (4)$$

where $\mathcal{B}$ is the set of related basis levels, $\mathcal{T}$ is the molecular training set, and $\mathcal{Q}$ contains the energy and property targets used during calibration. The quantity $q_{B,M}$ is the value obtained with basis level B for molecule M, q_M^{ref} is the corresponding reference value, the scale factors s_q make heterogeneous quantities comparable, the weights w_q control their relative importance, and γ weights the condition-number penalty. The logarithmic condition-number term is essential: diffuse primitives improve weak interactions and response properties, but unconstrained diffuse functions can create near-linear dependencies. In the MOLOPT construction, diffuse primitives are included within contractions, providing flexibility without making the overlap matrix numerically fragile.

Two aspects of this construction are important for the UZH protocol. The primitive exponents determine the radial completeness of each angular-momentum channel, whereas the contraction coefficients define compact functions that retain the flexibility needed for valence and polarization response. The condition-number penalty prevents an apparently improved molecular fit from becoming numerically unstable in condensed-phase GPW calculations. Thus, the basis optimization is not merely an enlargement of the basis; it is a constrained optimization of a transferable basis-set parameter set.

The UZH basis optimization follows this logic but makes the calibration explicit. Small molecules are used first because they probe chemically diverse bonding situations at modest cost, permit comparison to all-electron Gaussian references, and expose failures in geometries, dipoles, polarizabilities, and vibrational frequencies before the settings are transferred to solids. This molecular stage is not the final validation target; rather, it establishes a chemically balanced basis set that can then be assessed in a reproducible periodic workflow.

B. Separable dual-space Gaussian pseudopotentials

The GTH and HGH pseudopotentials are norm-conserving, angular-momentum-dependent pseudopotentials written in an analytic form that is compact in both real and reciprocal space [11, 12]. Their local part is

$$V_{\mathrm{loc}}^{\mathrm{pp}}(r) = -\frac{Z_{\mathrm{ion}}}{r} \operatorname{erf}\left(\alpha^{\mathrm{pp}} r\right) + \sum_{i=1}^{4} C_i^{\mathrm{pp}} \left(\sqrt{2}\alpha^{\mathrm{pp}} r\right)^{2i-2} \times \exp\left[-\left(\alpha^{\mathrm{pp}} r\right)^2\right], \quad (5)$$

with $\alpha^{\mathrm{pp}} = 1/(\sqrt{2} r_{\mathrm{loc}}^{\mathrm{pp}})$, where $r_{\mathrm{loc}}^{\mathrm{pp}}$ is the local pseudopotential radius. Here r is the electron–nucleus distance, Z_{ion} is the ionic charge, $i = 1, \ldots, 4$ enumerates the local Gaussian terms, and C_i^{pp} are their coefficients. The nonlocal part is written as

$$V_{\mathrm{nl}}^{\mathrm{pp}}(\mathbf{r}, \mathbf{r}') = \sum_{lm} \sum_{ij} \langle \mathbf{r} | p_i^{lm} \rangle h_{ij}^l \langle p_j^{lm} | \mathbf{r}' \rangle, \quad (6)$$

where l and m are angular-momentum and magnetic quantum numbers, i and j enumerate nonlocal projectors in a given l channel, p_i^{lm} are projector functions, and h_{ij}^l is the channel coupling matrix. The projectors have Gaussian radial factors,

$$\langle \mathbf{r}|p_i^{lm}\rangle = N_i^l Y^{lm}(\hat{r}) r^{l+2i-2} \exp\left[-\frac{1}{2}\left(\frac{r}{r_l}\right)^2\right]. \tag{7}$$

Here N_i^l is the projector normalization constant, Y^{lm} is the corresponding spherical harmonic, $\hat{r} = \mathbf{r}/r$, and r_l is the nonlocal projector radius for channel l. The nonlocal expression above is a Kleinman–Bylander-type separable representation [20]: each angular-momentum channel is represented by a small set of localized projectors and a low-dimensional coupling matrix h_{ij}^l rather than by a fully semilocal radial operator. This form makes the pseudopotential inexpensive to apply in large GPW calculations and directly usable in a plane-wave representation. Norm conservation requires the pseudovalence states to reproduce the all-electron valence norm and scattering properties outside the core region for each angular-momentum channel. Together with the use of more than one projector per channel, this constraint improves transferability across different bonding environments. The Gaussian radial factors are central to the dual-space construction because they lead to analytic Fourier transforms, so the same GTH parameter set can be evaluated consistently in CP2K/Quickstep and in SIRIUS.

A natural extension within the same GTH family is a nonlinear core correction (NLCC), in which a smooth partial core charge is included in the exchange-correlation functional to account for core-valence density overlap. GTH-NLCC potentials have been constructed by adding a compact Gaussian core charge to the dual-space form and can improve thermochemical transferability, especially for spin-polarized atoms, radical chemistry, and functional transfer [21, 22]. We therefore regard NLCC as a compatible extension of the UZH loop: if the SIRIUS-GTH-UZH versus SIRIUS-FP-LAPW comparison identifies a residual pseudopotential error with a core-valence origin, the CP2K ATOM refit can include NLCC parameters in addition to the standard GTH radii, local coefficients, and projector couplings. By contrast, ultrasoft pseudopotential and projector augmented-wave datasets are important alternative frozen-core formalisms in plane-wave electronic-structure codes [23, 24], but they are not part of the present decomposition because they would change the pseudopotential representation rather than improve the CP2K GTH/MOLOPT parameter files.

This representation is well suited to GPW calculations because it can be evaluated efficiently with Gaussian orbitals and on real-space grids, while also being compatible with plane-wave calculations. The relativistic HGH generalization extends the construction across the periodic table and retains the separable analytic structure [12]. In the UZH protocol, the GTH form is optimized with CP2K's ATOM program against DKH3 atomic reference calculations, including valence and semicore information as needed.

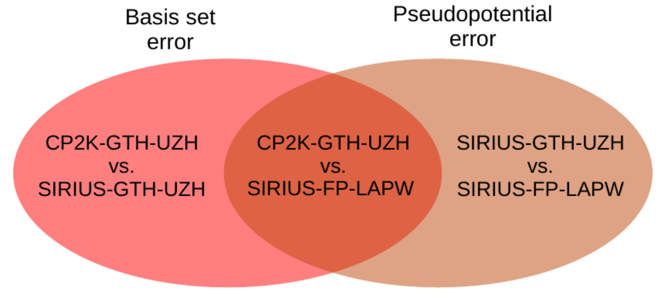

FIG. 1. Error decomposition used in the UZH protocol. CP2K-GTH-UZH compared with SIRIUS-FP-LAPW gives the total practical error. CP2K-GTH-UZH compared with SIRIUS-GTH-UZH isolates the Gaussian-basis contribution. SIRIUS-GTH-UZH compared with SIRIUS-FP-LAPW isolates the pseudopotential contribution.

C. Error decomposition

Let $E_{\mathrm{C}}^{\mathrm{GTH}}(V)$ denote CP2K-GTH-UZH energies, $E_{\mathrm{S}}^{\mathrm{GTH}}(V)$ SIRIUS-GTH-UZH energies, and $E_{\mathrm{S}}^{\mathrm{AE}}(V)$ SIRIUS-FP-LAPW all-electron energies for the same structure and volume, respectively. The total deviation of the practical CP2K protocol is

$$\Delta_{\mathrm{tot}}(V) = E_{\mathrm{C}}^{\mathrm{GTH}}(V) - E_{\mathrm{S}}^{\mathrm{AE}}(V). \tag{8}$$

It can be decomposed as

$$\Delta_{\mathrm{tot}}(V) = \Delta_{\mathrm{basis}}(V) + \Delta_{\mathrm{pp}}(V), \tag{9a}$$
$$\Delta_{\mathrm{basis}}(V) = E_{\mathrm{C}}^{\mathrm{GTH}}(V) - E_{\mathrm{S}}^{\mathrm{GTH}}(V), \tag{9b}$$
$$\Delta_{\mathrm{pp}}(V) = E_{\mathrm{S}}^{\mathrm{GTH}}(V) - E_{\mathrm{S}}^{\mathrm{AE}}(V). \tag{9c}$$

The first term measures the CP2K Gaussian-basis error for a fixed pseudopotential; the second term measures the pseudopotential error in a systematic basis. In practice, the comparisons are performed not at a single volume but over equation-of-state curves. We use the ε metric of the verification project [6],

$$\varepsilon(a,b) = \sqrt{\frac{\sum_i \left[E_a(V_i) - E_b(V_i)\right]^2}{\sqrt{\sum_i \left[E_a(V_i) - \langle E_a\rangle\right]^2 \sum_i \left[E_b(V_i) - \langle E_b\rangle\right]^2}}}, \tag{10}$$

where a and b label the two methods being compared, i runs over the volume grid, the volumes V_i are sampled around the equilibrium region, and $\langle E_a\rangle$ and $\langle E_b\rangle$ are the corresponding mean energies over that grid. This dimensionless metric is sensitive to equation-of-state differences while remaining comparable across chemically distinct elements.

Figure 1 summarizes this diagnostic construction and shows which pairwise comparison yields the total, Gaussian-basis, and pseudopotential contributions.

III. COMPUTATIONAL METHODS

A. Molecular calibration

The molecular calibration follows the MOLOPT strategy of optimizing transferable, numerically well-conditioned Gaussian basis sets for chemical accuracy across molecules and condensed phases [10]. We benchmarked double-zeta valence plus polarization (DZVP), triple-zeta valence plus polarization (TZVP), and triple-zeta valence plus two polarization shells (TZV2P) UZH-MOLOPT basis sets on a database of small molecules containing diverse bonding motifs; the corresponding CP2K molecular test calculations are archived in the MolTest Zenodo dataset [25]. Reference calculations were performed with Gaussian 16 [26] using all-electron def2 quadruple-zeta valence plus polarization (def2-QZVP) settings [27], tight self-consistent-field convergence, and dense numerical integration grids. The CP2K calculations paired all-electron Gaussian-and-augmented-plane-wave (GAPW) calculations [28] with matching GTH-UZH pseudopotential setups to separate basis and pseudopotential effects at the molecular level. We evaluated relative bond-length errors, maximum angle and torsional errors, dipole moments, average polarizabilities, and vibrational frequencies. Structures with incomplete reference or CP2K convergence and structures identified by the frequency analysis as non-minima were excluded from the statistics. Additional details and property-specific statistics are given in the Supplemental Material [29]. Specifically, Sec. I of the Supplemental Material documents the CP2K ATOM, basis-optimization, and SIRIUS setup details needed to reproduce the workflow, while Supplemental Material Figs. 1–5 collect the molecular calibration statistics that are too extensive for the main text, including angle errors, dihedral errors, dipole moments, bond-class trends, and the Gaussian 16/CP2K all-electron reference comparison.

B. Periodic verification workflow

Periodic benchmarks were generated with the unary-crystal verification workflow of AiiDA common workflows [5, 6]. For each element, face-centered cubic, body-centered cubic, simple cubic, and diamond prototypes were considered where applicable. Seven volumes were sampled from 94% to 106% around the reference volume and fitted to a Birch-Murnaghan equation of state. The same structural inputs and workflow logic were used for CP2K-GTH-UZH, SIRIUS-GTH-UZH, and SIRIUS-FP-LAPW.

CP2K/Quickstep calculations used the GPW method with UZH-MOLOPT basis sets and the corresponding GTH-UZH pseudopotentials. SIRIUS-GTH-UZH calculations used the same GTH-UZH pseudopotentials in a plane-wave representation, thereby removing the Gaussian-basis approximation while retaining the pseudopotential. SIRIUS-FP-LAPW calculations served as the all-electron reference. SIRIUS is a domain-specific electronic-structure library that implements pseudopotential plane-wave and full-potential linearized augmented-plane-wave methods and is designed for parallel and accelerator-enabled calculations [14–16]. In the FP-LAPW mode, the unit cell is partitioned into atom-centered augmentation spheres and an interstitial region; core and valence electrons are treated explicitly, and the density and effective potential are not constrained to a shape approximation. This makes SIRIUS-FP-LAPW an appropriate reference for identifying the residual core approximation in the pseudopotential calculations.

The relativistic treatment was kept consistent with the scalar-relativistic GTH/HGH construction. In the all-electron SIRIUS reference, the full-potential species definitions specify the relativistic treatment of core and valence states; for the FP-LAPW benchmarks reported here, valence scalar relativity is treated with the infinite-order regular approximation (IORA) [30]. This IORA setting is part of the all-electron reference definition, while in the pseudopotential calculations the corresponding DKH3 atomic-reference information is folded into the GTH-UZH potential. Spin-orbit effects are therefore not part of the error decomposition considered here. This choice is deliberate: the objective is to separate Gaussian-basis incompleteness from scalar-relativistic pseudopotential transferability for the CP2K settings used in production GPW calculations. The availability of SIRIUS inside CP2K makes this decomposition especially useful because CP2K can be used both as the production GPW code and as an interface to a systematic plane-wave or all-electron reference. Since SIRIUS is a CP2K component, the verification, optimization, and subsequent improvement loop can be carried out holistically within the CP2K ecosystem rather than by combining unrelated codes.

C. CP2K ATOM and closed-loop optimization

The optimization loop uses the updated CP2K ATOM code developed as part of the present UZH work. Inspection of the CP2K source and input reference shows that the ATOM input layer supports energy calculations, basis optimization, and pseudopotential optimization as distinct run types. For the present UZH fits, the all-electron atomic references used in the pseudopotential optimization were generated with the DKH3 Hamiltonian. The input layer can read, print, fit, and export GTH pseudopotentials and also supports other scalar-relativistic options, including zeroth-order regular approximation and Douglas–Kroll–Hess variants [7, 8]. The basis optimizer uses template, work, and output basis files and allows the condition number of the overlap matrix to enter the training objective. The pseudopotential optimizer constructs all-electron and pseudopotential atomic references with explicit core, valence, and semicore occupations, evaluates orbital energies and radial charges, and uses a Powell-type derivative-free optimization to fit the compact GTH parameter set [31]. These implementation details are important because they make CP2K not merely the code under test, but also the code that can generate revised basis and scalar-relativistic pseudopotential parameter files after the error decomposition has identified the limiting approximation.

IV. RESULTS AND DISCUSSION

The results are organized as a diagnostic sequence that follows the order of the approximations in the protocol. Section IV.A identifies the smallest converged UZH-MOLOPT basis set suitable for production GPW calculations while retaining numerical stability. Section IV.B then quantifies the total error of the original CP2K-GTH-UZH settings before optimization. Section IV.C separates this total error into Gaussian-basis and pseudopotential contributions by inserting the SIRIUS-GTH-UZH reference between CP2K-GTH-UZH and SIRIUS-FP-LAPW. Finally, Sec. IV.D evaluates the optimized UZH settings in the same order and shows whether the targeted basis-set and pseudopotential revisions reduce the dominant outliers.

A. Molecular basis calibration

The molecular benchmark confirms the expected systematic behavior of the UZH-MOLOPT sequence. Figure 2 reports the first and most local test: relative bond-length errors with respect to the all-electron molecular reference. The distribution narrows from DZVP to TZVP and TZV2P, with the median relative error remaining small across the database. Carbon-containing bonds are particularly stable, while O- and F-containing compounds are more sensitive to polarization and diffuse flexibility in the smaller basis sets. The TZV2P level removes most of these systematic trends.

This molecular step is not intended to replace the periodic verification workflow; it is the chemically local calibration that makes the subsequent solid-state test meaningful. A basis that performs well for equation-of-state curves but gives unbalanced molecular geometries, dipoles, or polarizabilities would not be transferable. Conversely, a molecularly optimized basis still has to be checked in condensed phases because diffuse functions, near-linear dependencies, and grid-coupling errors can become more severe in periodic calculations. The UZH protocol therefore uses the molecular benchmark as a first filter: the basis sequence must improve monotonically for local structural observables while keeping the overlap matrix sufficiently well conditioned for production GPW calculations.

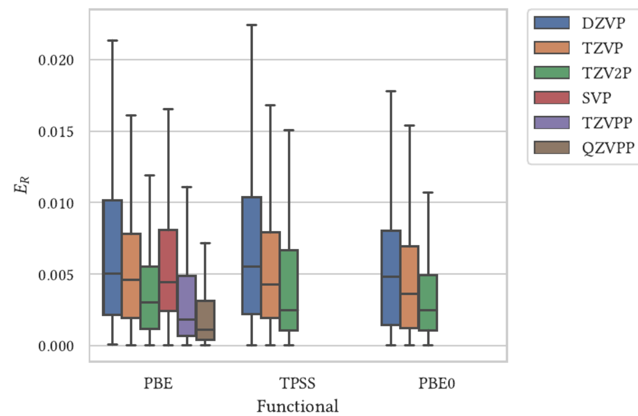

FIG. 2. Distribution of relative bond-length errors for the molecular calibration set. Outliers are omitted from the box plot to emphasize the median and interquartile behavior. The systematic narrowing from DZVP to TZV2P indicates controlled basis-set convergence for local structural properties.

Nonlinear bond angles and dihedral angles extend the same analysis beyond local bond distances. They are more sensitive than bond lengths to subtle changes in the electronic structure and to the automated matching of nearly symmetric structures, but the hierarchy of basis quality remains visible; Supplemental Material Figs. 1 and 2 report the full angle and dihedral-error distributions, respectively.

The response-property tests complete the molecular calibration. Dipole moments follow the same systematic basis-set trend with the full distribution given in Supplemental Material Fig. 3. Figure 3 shows the average molecular polarizability, the most demanding response observable in this set. The larger variance observed for polarizabilities is consistent with the need for diffuse flexibility, but the absence of catastrophic outliers at the TZV2P level indicates that the condition-number constraint in Eq. (4) is effective. The supplemental bond-class and reference-code comparisons in Supplemental Material Figs. 4 and 5 further support this choice: they show that the TZV2P improvement is not confined to the aggregate bond-length distribution and that the molecular calibration is not dominated by a single reference-code mismatch.

Together, the bond-length and polarizability benchmarks, complemented by the angular, dihedral, and dipole-moment statistics in Supplemental Material Figs. 1–3, motivate two practical choices. First, DZVP is useful as an economical screening level, but it should not be interpreted as a converged basis for response-sensitive properties. Second, TZV2P provides a more balanced reference level for the UZH optimization loop because it improves both covalent and polar bonding motifs without introducing the numerical fragility that would compromise large periodic calculations. This balance between chemical flexibility and numerical stability is essential:

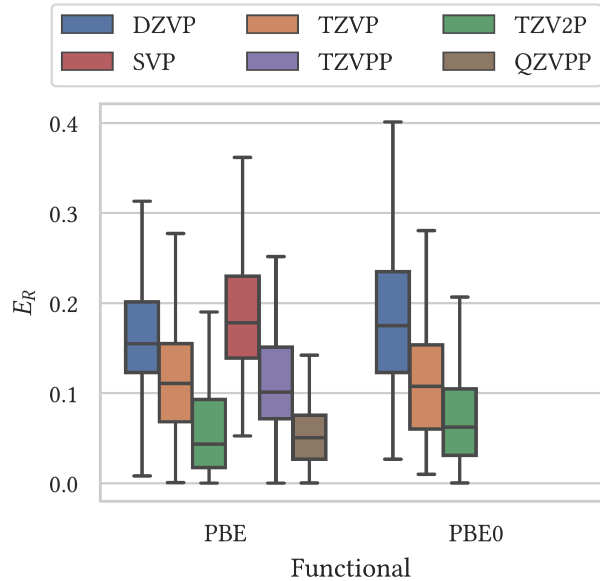

FIG. 3. Distribution of relative errors for average molecular polarizabilities. Polarizabilities are the most demanding molecular response test in this set because they require sufficient diffuse flexibility; meta-generalized-gradient data are omitted because this property was not available for the corresponding CP2K calculations.

the goal is not to make the largest possible Gaussian basis, but to construct a basis that can be improved systematically together with the corresponding GTH-UZH pseudopotential files. Consequently, all periodic CP2K-GTH-UZH calculations discussed below use the TZV2P UZH-MOLOPT basis level unless stated otherwise.

B. Total error before and after improvement

Figure 4 compares the total practical error of the original and improved CP2K-GTH-UZH settings relative to the SIRIUS-FP-LAPW reference. In the original settings, several elements exhibit $\varepsilon > 1$ for at least two unary structures, including noble gases, selected transition metals, and heavier main-group elements. After applying the UZH protocol, the corresponding CP2K-GTH-UZH-new map is substantially cleaner, with the largest residual deviations confined to a smaller set of chemically interpretable cases. This before–after comparison is the error a user would observe if CP2K were treated as a black box. It is therefore the most direct measure of the protocol's practical accuracy, but by itself it does not reveal the origin of the discrepancy.

The important point is that the outliers do not have a single chemical or numerical origin. Noble-gas solids are especially sensitive to the radial flexibility and diffuseness of the valence basis because their equation-of-state curves are shallow and dominated by weak interactions. Several transition-metal cases, in contrast, point to the quality of the frozen-core representation, semicore treatment, and the scalar-relativistic transferability encoded in the DKH3-fitted pseudopotential. Heavy main-group elements can combine both effects. The total-error map is therefore deliberately used as a starting point rather than as a final ranking of elements: it identifies where the original settings are insufficient, but the remedial action requires the decomposition in the next subsection.

This distinction is central for interpreting ε. A large total error does not automatically mean that the CP2K Gaussian basis must be enlarged, and a small total error does not prove that basis and pseudopotential errors are individually negligible; partial cancellation is possible. For parameter development, the total CP2K-GTH-UZH versus SIRIUS-FP-LAPW comparison in Fig. 4 has to be read together with the two intermediate comparisons summarized in Table I. Only then can the protocol decide whether the next iteration should modify the MOLOPT basis, refit the GTH pseudopotential parameters, or treat both components.

C. Separating basis and pseudopotential errors

The decomposition in Eq. (9) reveals three distinct regimes, which are shown in Fig. 5 from top to bottom as basis-set and pseudopotential errors. First, the noble gases and selected elements such as Ba are basis limited. SIRIUS-GTH-UZH agrees well with SIRIUS-FP-LAPW, indicating that the GTH-UZH pseudopotential is transferable for these tests, but CP2K-GTH-UZH deviates from SIRIUS-GTH-UZH. These cases point to missing diffuse or polarization flexibility in the Gaussian basis for weakly bound or highly compressible solids. This observation is physically plausible because noble-gas equation-of-state curves are shallow and therefore sensitive to small basis-set and basis-set-superposition effects.

Second, for elements such as Ca, Cr, Pd, Rh, Ru, and Tc, SIRIUS-GTH-UZH differs substantially from SIRIUS-FP-LAPW, while CP2K-GTH-UZH remains close to SIRIUS-GTH-UZH. These cases are pseudopotential limited. Improving the Gaussian basis alone would not remove the dominant error because the same discrepancy is already present in the plane-wave representation of the pseudopotential.

Third, some heavy elements show mixed behavior. In these cases both pairwise comparisons are non-negligible, and the improvement strategy must combine basis-set extension or reoptimization with pseudopotential refitting.

D. Improved UZH settings

The classification in Table I was used to revise the settings in a targeted way. Basis-limited elements were treated by extending or reoptimizing the UZH-MOLOPT basis while retaining the condition-number penalty that

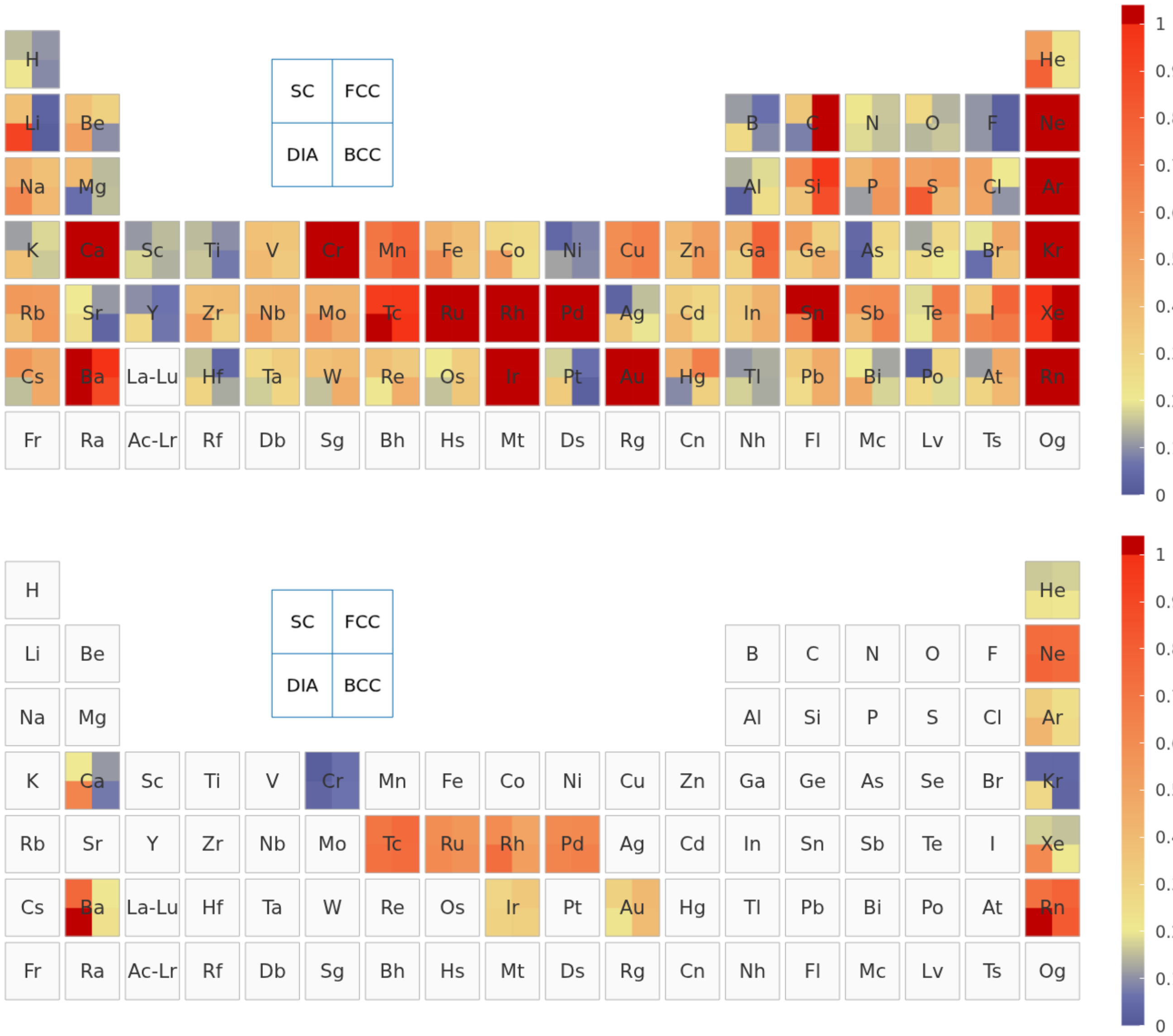

FIG. 4. Total equation-of-state error relative to SIRIUS-FP-LAPW, expressed with the ε metric of Eq. (10). Top: original CP2K-GTH-UZH settings. Bottom: improved CP2K-GTH-UZH-new settings. Each map contains both Gaussian-basis and pseudopotential errors and therefore represents the practical production-level CP2K error before and after applying the UZH protocol.

TABLE I. Qualitative classification of dominant error sources in the original UZH settings. The classification is based on the pairwise comparisons in Figs. 4 and 5.

Dominant source	Diagnostic signature	Representative elements
Basis set	Large CP2K-GTH-UZH vs SIRIUS-GTH-UZH error	He, Ne, Ar, Kr, Xe, Rn, Ba
Pseudopotential	Large SIRIUS-GTH-UZH vs SIRIUS-FP-LAPW error	Ca, Cr, Pd, Rh, Ru, Tc
Mixed	Both components visible	Hg and selected heavy elements

prevents unstable diffuse functions. Pseudopotential-limited elements were treated by refitting the GTH parameters with the CP2K ATOM code, including DKH3 atomic reference information and semicore states where appropriate. This targeted route differs from a brute-force convergence exercise: it changes only the component that the error decomposition identifies as limiting.

Figure 6 reports the two optimized comparisons that define the remaining error components. The top panel compares CP2K-GTH-UZH-new with SIRIUS-

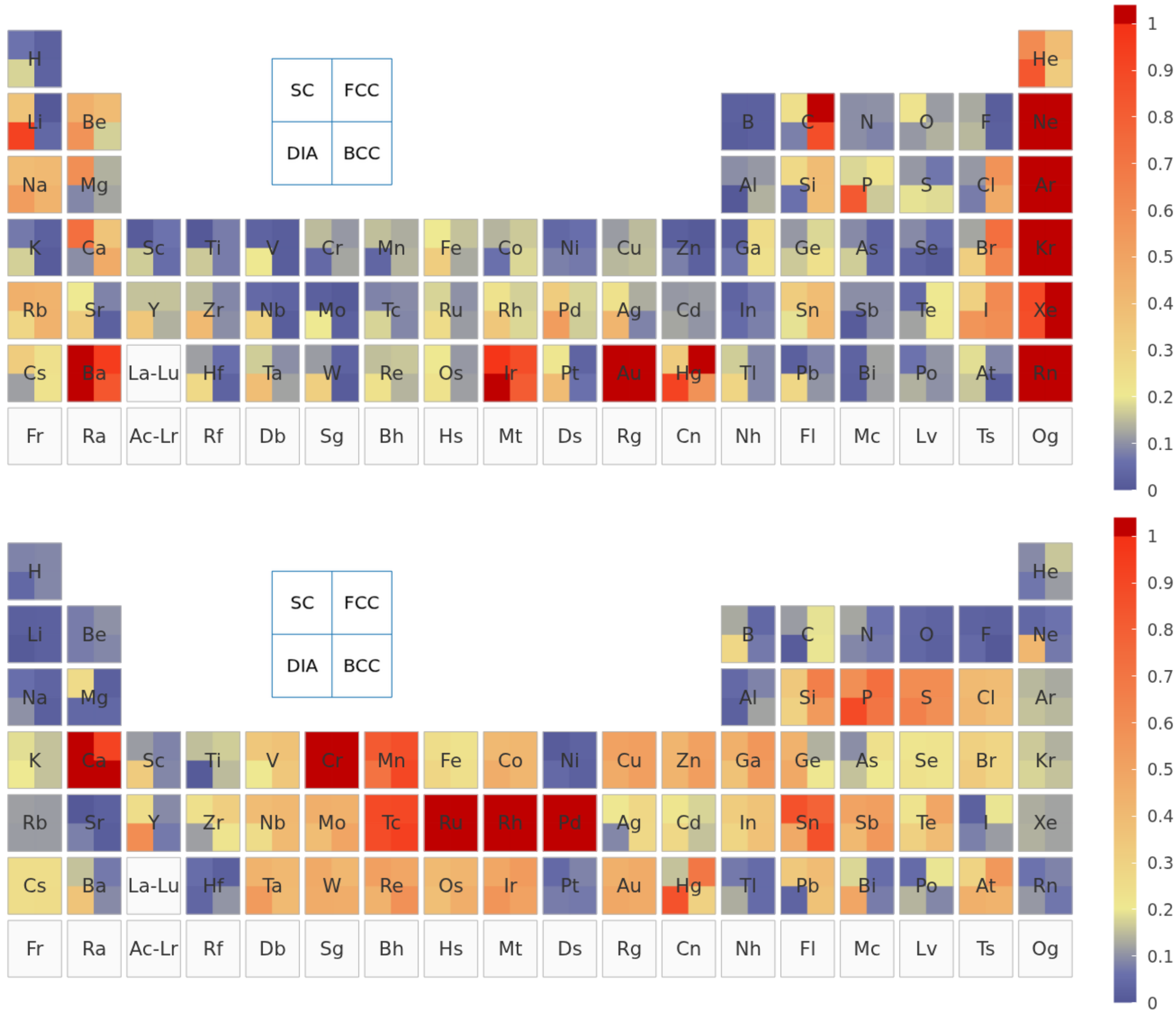

FIG. 5. Error decomposition for the original UZH settings. Top: CP2K-GTH-UZH relative to SIRIUS-GTH-UZH, which isolates the Gaussian-basis error. Bottom: SIRIUS-GTH-UZH relative to SIRIUS-FP-LAPW, which isolates the pseudopotential error.

GTH-UZH-new and therefore gives the Gaussian-basis contribution for the optimized settings. The bottom panel compares SIRIUS-GTH-UZH-new with SIRIUS-FP-LAPW and gives the pseudopotential contribution. At the level of the underlying signed equation-of-state energy differences, the superposition of the upper and lower comparisons corresponds to the remaining total CP2K-GTH-UZH-new error relative to SIRIUS-FP-LAPW shown in the bottom panel of Fig. 4; the positive ε values themselves should therefore be interpreted as component norms rather than added element by element. Read together with the total-error reduction in Fig. 4, these maps show that the improvement is not a cosmetic redistribution of errors but a targeted reduction of the components identified as limiting. Supplemental Material Fig. 6 places the residual pseudopotential map next to the direct pseudopotential-update map, making clear which refits reduce the all-electron discrepancy and which elements retain a pseudopotential-limited contribution. Supplemental Material Figs. 7–11 then show representative equation-of-state curves for Cr, Ba, Ne, Au, and Ga, connecting the periodic-table heat-map colors to the underlying energy-volume data and showing how basis-limited, pseudopotential-limited, and mixed cases appear at the curve level.

The improved settings reduce the dominant outliers of the original protocol and make the remaining discrepancies more interpretable. In particular, noble-gas and Ba errors are reduced by basis-set changes, while the transition-metal cases respond to pseudopotential refitting. The resulting UZH protocol is therefore not a fixed table of parameters but a reproducible recipe for diag-

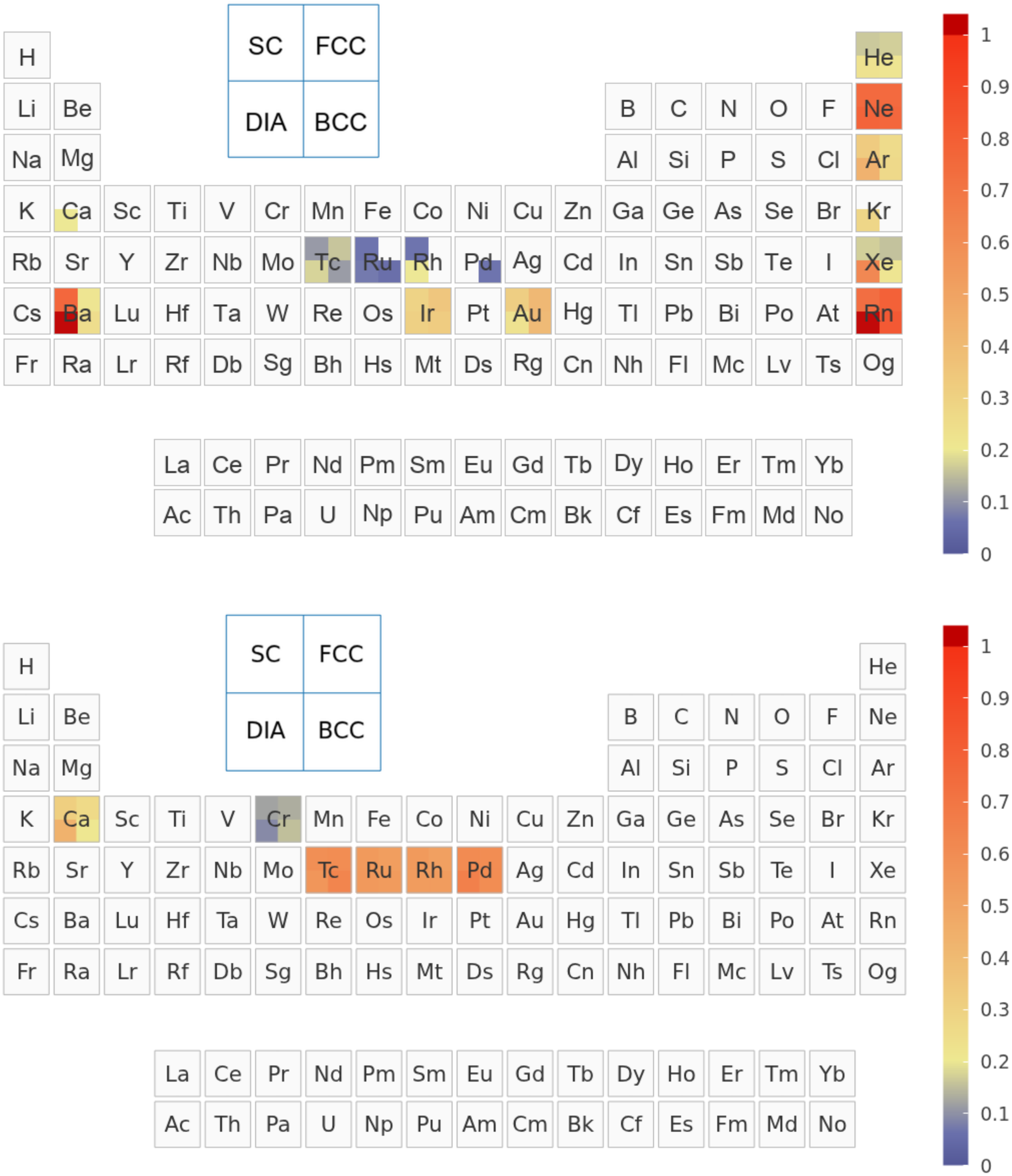

FIG. 6. Optimized UZH error decomposition. Top: CP2K-GTH-UZH-new relative to SIRIUS-GTH-UZH-new, i.e., the Gaussian-basis component after optimization. Bottom: SIRIUS-GTH-UZH-new relative to SIRIUS-FP-LAPW, i.e., the pseudopotential component after optimization.

nosing, optimizing, and validating CP2K settings.

V. IMPLICATIONS FOR CP2K VERIFICATION

The UZH protocol turns CP2K verification into a diagnostic and constructive workflow. The comparison with SIRIUS-FP-LAPW gives the total accuracy relevant for production calculations. The intermediate SIRIUS-GTH-UZH calculation tells whether the Gaussian basis or the pseudopotential is responsible. The newly developed CP2K ATOM machinery then supplies the internal route for improvement: basis optimization with condition-number control, DKH3 atomic references, and GTH parameter fitting. This is particularly valuable for users who need reliable calculations across molecules and solids, because the protocol connects molecular transferability to periodic equation-of-state precision.

The protocol also clarifies what should be considered converged. Increasing the GPW grid cutoff alone cannot remove a Gaussian-basis error. Likewise, replacing a basis set cannot repair a pseudopotential whose scattering or semicore transferability is insufficient. Conversely, once the SIRIUS-GTH-UZH and SIRIUS-FP-LAPW curves agree, the remaining CP2K discrepancy is a basis issue that can be addressed without changing the pseudopotential. This separation is the practical advantage of combining CP2K and SIRIUS in a single AiiDA-based verification framework.

VI. CONCLUSIONS

We have presented the UZH protocol for constructing, diagnosing, and improving CP2K basis-set and pseudopotential settings. The protocol combines molecularly optimized MOLOPT basis sets with GTH separable dual-space Gaussian pseudopotentials, calibrates the basis on small molecules, and validates the resulting settings through AiiDA unary-crystal equation-of-state workflows. By comparing CP2K-GTH-UZH, SIRIUS-GTH-UZH, and SIRIUS-FP-LAPW, the total CP2K error can be decomposed into Gaussian-basis and pseudopotential components. The same CP2K infrastructure used for production calculations also provides the newly developed ATOM and basis-optimization tools required to improve the limiting component. The resulting workflow provides a reproducible path toward systematically improvable CP2K simulations across molecules and condensed phases. The present UZH release provides MOLOPT Gaussian basis sets and GTH pseudopotential parameter files for three exchange-correlation levels: Perdew–Burke–Ernzerhof (PBE) at the generalized-gradient-approximation level, strongly constrained and appropriately normed (SCAN) at the meta-generalized-gradient level, and Perdew–Burke–Ernzerhof hybrid (PBE0) as a hybrid density-functional approximation with exact exchange [32–34]. We therefore propose the UZH protocol as a new standard for CP2K pseudopotential calculations and for the corresponding all-electron verification calculations, including CP2K's GAPW approximation; the all-electron branch of the protocol and the accuracy and scope of the GAPW approximation will be discussed in detail in future work.

ACKNOWLEDGMENTS

The authors gratefully acknowledge the computing time made available to them on the high-performance computer OTUS at the Paderborn Center for Parallel Computing (PC2), a National High-Performance Computing (NHR) Center. PC2 is jointly supported by the Federal Ministry of Research, Technology and Space and the state governments participating in the NHR joint funding program (`www.nhr-verein.de/en/our-partners`).

DATA AVAILABILITY

The curated release repository for the UZH protocol is available at `https://github.com/DCM-Uni-Paderborn/UZH-protocol`. It contains the optimized CP2K basis and pseudopotential files, manuscript figures, provenance manifests, and the release structure for the CP2K-GTH-UZH, SIRIUS-GTH-UZH, and SIRIUS-FP-LAPW workflow data. Complete workflow inputs, parsed outputs, and post-processing scripts will be deposited in the same repository before final publication. Section V of the Supplemental Material mirrors this organization and lists the intended release layout so that plotted figures, workflow records, and optimized parameter files can be traced to the deposited data.

AUTHOR CONTRIBUTIONS

Hossein Mirhosseini: conceptualization, data curation, formal analysis, methodology, software, visualization, writing–original draft, writing–review and editing. Tiziano M. A. Müller: data curation, methodology, software, validation, writing–review and editing. Matthias Krack: methodology, basis-set and pseudopotential curation, validation, writing–review and editing. Thomas D. Kühne: conceptualization, methodology, supervision, funding acquisition, writing–original draft, writing–review and editing. Jürg Hutter: conceptualization, methodology, supervision, writing–review and editing.

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